

[Length check] SPR-IRL Method and Experiments

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1 Method: SPR-IRL

This section defines Symmetric Preference Recovery (SPR-IRL). SPR-IRL is not a single model but a procedure comprising (i) symmetric estimation over shared features, (ii) out-of-sample stability assessment, and (iii) a contrast against sign expectations fixed in advance.

1.1 Continuous Action and Symmetric Estimation

Let the investor types be $\mathcal{I} = \{\text{foreign, institutional, retail}\}$. Writing $V_{i,t}^{buy}$ and $V_{i,t}^{sell}$ for the daily gross buy and sell value of type i , the action is

$$u_{i,t} = \frac{V_{i,t}^{buy} - V_{i,t}^{sell}}{V_{i,t}^{buy} + V_{i,t}^{sell}} \in [-1, 1]. \quad (1)$$

Because the denominator is the type's own gross turnover rather than market-wide turnover, $u_{i,t}$ is a common scale unaffected by differences in trading size across types, and unlike a binary label it preserves intensity.

All three types receive the same state features $x_{i,t} \in \mathbb{R}^3$, the same context $C_t \in \mathbb{R}^2$, and the same model form; only the parameters $\theta_i = (\beta_i, B_i, \alpha_i)$ are estimated separately on each type's own data. Since no type is assigned a different feature a priori, any difference in the results is learned rather than imposed, and this is what licenses direct comparison of coefficients across types.

The training sample is $(x_{i,t}, C_t, a_{i,t+1})$ with $a_{i,t+1} = u_{i,t+1}$: a day- t state explains day- $t+1$ behaviour, so no future trade enters the features.

1.2 Reward Features and Market Context

All three features and both contexts are standardized on the training indices of each split only, so every coefficient reads as the effect of a one-standard-deviation move.

(F1) Medium-term momentum $x_t^{mom} = \log(P_t/P_{t-20})$. The 20-day window is pre-specified for interpretability and was not tuned on the data. A positive coefficient indicates trend following and a negative one contrarian trading; the sign expectation follows the literature on type-specific momentum trading [1, 3].

(F2) Herding is the cross-sectional average of the previous day's actions of the two types other than i ,

$$x_{i,t}^{herd} = \frac{1}{2} (u_{j,t-1} + u_{k,t-1}), \quad j, k \neq i. \quad (2)$$

The one-day lag is not an information constraint but a design choice that keeps the same-day accounting offset between types out of the predictor set (§2.3). The sign expectation

follows the institutional herding literature [6], which however measures herding *within* the institutional group whereas Equation (2) measures a relation *between* types.

(F3) Underwater gap uses an average-cost proxy $\bar{P}_{i,t}$ and the day's gross-trade VWAP $P_{i,t}^{vwap}$,

$$x_{i,t}^{uw} = \max\left(\frac{\bar{P}_{i,t} - P_{i,t}^{vwap}}{\bar{P}_{i,t}}, 0\right). \quad (3)$$

Truncation at zero makes the variable reference-point dependent by construction. The average cost recursively updates an effective inventory from gross quantities and values, multiplying the previous day's inventory by a forgetting factor $\rho = 0.98$ (half-life about 34 trading days). The sign expectation follows the disposition-effect literature [7, 8]. Because the proxy is built from aggregate flow, the coefficient on x^{uw} cannot be read as a direct test of that effect.

The context consists of the one-day KOSPI200 return r_t and the KRW/USD level $z_t = (\log e_t - \mu_{t-1,252})/\sigma_{t-1,252}$, where the mean and standard deviation use only data through $t-1$. Because Samsung Electronics carries a large KOSPI200 weight, stock momentum overlaps the market return and feature-context correlations range from -0.29 to 0.22 ; individual coefficients are therefore conditional associations given the other terms.

1.3 Myopic Reward and the Reduction to Lasso

Define the action score of type i as

$$q_{i,t} = (\beta_i + B_i C_t)^\top x_{i,t} + \alpha_i^\top C_t. \quad (4)$$

Here α is a level effect that shifts behaviour even at average features and B a slope effect that changes feature sensitivity across contexts. Including interactions while omitting the lower-order term C_t is a standard specification error, so the inclusion of α is not something to justify empirically. With the one-step reward $R_i = q_{i,t}a - \frac{\kappa}{2}a^2$ the optimal policy is

$$a_{i,t+1}^* = \text{clip}(q_{i,t}/\kappa, -1, 1). \quad (5)$$

Without the quadratic term the solution would be ± 1 almost always and intensity could not be reproduced. We fix $\kappa = 1$ for identification, which normalizes the units of the reward and the coefficients rather than asserting that the empirical curvature is one. The training objective is

$$\min_{\theta_i} \frac{1}{|\mathcal{D}_i|} \sum_{t \in \mathcal{D}_i} (a_{i,t+1}^* - a_{i,t+1})^2 + \lambda_i \|\theta_i\|_1, \quad (6)$$

optimized independently per type with Adam [5].

PROPOSITION 1.1 (REDUCTION TO LASSO). *Let $z_{i,t} = [x_{i,t}; \text{vec}(x_{i,t}C_t^\top); C_t] \in \mathbb{R}^{11}$ and $\theta_i = [\beta_i; \text{vec}(B_i); \alpha_i]$, so*

that $q_{i,t} = \theta_i^\top z_{i,t}$. If $|q_{i,t}| \leq 1$ throughout the training sample, the clip in Equation (5) never binds, $a^* = q$, and Equation (6) becomes exactly an L_1 -penalized (Lasso) regression of $a_{i,t+1}$ on $z_{i,t}$.

In our sample the maximum $|q|$ is 0.842 for foreign, 0.587 for retail and 0.135 for institutional investors, and the saturation rate is exactly zero in all 45 splits, so the premise holds. In SPR-IRL, “recovering a preference” and “regressing order flow” are therefore numerically indistinguishable.

This is not a defect but a consequence of myopia. If the agent looks one step ahead, today’s action does not affect anything beyond tomorrow, so what the agent wants (the reward) collapses onto what it does (the policy). Applying the stochastic policy of maximum-entropy IRL [9], $\pi \propto \exp(R/\tau)$, makes π a truncated normal with mean q whose point estimate coincides with the above; the reduction is thus a property of the myopic, quadratic-reward setting in general. The value of the IRL formulation lies in the extension path: once state dynamics and multi-period value are introduced, the reward separates from the policy and yields a structural recovery that regression cannot reach (§3).

1.4 Validation Protocol

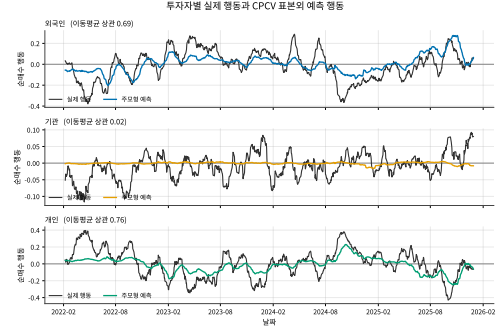
Rather than reporting the recovered coefficients as they are, we apply four stages. **(V1)** Combinatorial purged cross-validation [2] with ten temporal folds of which two are test folds gives $\binom{10}{2} = 45$ splits, with a one-day purge, a five-day embargo, and all standardization fitted on training indices only. Because splits share observations, sign consistency is interpreted as a stability indicator, not a p -value. **(V2)** To address that limitation we partition the sample into calendar-month blocks, resample 1,000 times, and obtain 95% confidence intervals; month blocks preserve the autocorrelation of daily flow within each block. **(V3)** Because CPCV can place a test fold before a training fold, we add an expanding walk-forward: train on the first 400 observations, predict the next 60 trading days after a five-day purge, and repeat while expanding by 60 days, for ten windows. **(V4)** To confirm the results are not an artefact of the L_1 penalty we re-estimate under L_2 and no penalty and compare signs.

Metrics are MAE, RMSE, directional accuracy (the fraction with $\text{sign}(a^*) = \text{sign}(a)$), the actual–predicted correlation, and the saturation rate. Because action standard deviations differ across types, absolute errors are not compared across types. As a naive reference we report a persistence rule that predicts the previous day’s own action.

2 Experiments

2.1 Data

From daily records for Samsung Electronics (005930) we use the most recent 973 rows for which the 252-day exchange-rate history, the 20-day momentum, and the next day’s action can all be computed; state dates run 2022-01-06 to 2025-12-29. The single-stock design is not a sample limitation but a controlled setting that removes stock fixed effects



각 날짜마다 실제 CPCV 시뮬레이션을 통해 평균한 뒤 실제값과 예측값에 20일 이동평균을 적용했다. 예측 잔차가 실재보다 작은 것은 L1 정규화에 의한 특수 제로값이다. 플롯 안 상단은 이동평균 계층 간 값이며, 실제 행동의 상단(보 2)보다 높다.

Figure 1: Actual and out-of-sample predicted net buying (20-day moving average). The nine CPCV test predictions per date are averaged first. Correlations in the titles are between the smoothed series and are not directly comparable to the daily correlations in Table 1.

Table 1: Out-of-sample performance under CPCV (mean $\pm$ s.d. over 45 splits) and the persistence baseline

Metric	Foreign	Inst.	Retail
<i>SPR-IRL</i>			
Directional accuracy	0.636 $\pm$ 0.039	0.542 $\pm$ 0.038	0.608 $\pm$ 0.044
Correlation	0.278 $\pm$ 0.110	0.068 $\pm$ 0.064	0.245 $\pm$ 0.118
MAE	0.195 $\pm$ 0.019	0.094 $\pm$ 0.008	0.264 $\pm$ 0.029
MAE / action s.d.	0.759	0.756	0.791
Saturation rate	0.000	0.000	0.000
<i>Persistence ($a_{t+1} \leftarrow a_t$)</i>			
Directional accuracy	0.642	0.547	0.616
Correlation	0.359	0.136	0.311

and liquidity differences and isolates preference heterogeneity across types.

Mean actions are near zero for all three types, but standard deviations differ (retail 0.334, foreign 0.257, institutional 0.125) and first-order autocorrelations are 0.345, 0.401 and 0.149. Same-day actions across types are strongly negatively related (foreign–retail -0.772 , institutional–retail -0.499), reflecting the market-clearing constraint that one type’s buying must appear as another’s selling; this bounds the interpretation in §2.3.

Training settings are, per type, foreign 75 epochs / batch 256 / lr 0.001 / $\lambda = 0$; institutional 10 / 2048 / 0.0005 / 0.01; retail 20 / 512 / 0.001 / 0.0003, with seed 42, $\kappa = 1$, and 11 parameters per type (β 3, B 6, α 2).

2.2 Out-of-Sample Behaviour Reproduction

In Figure 1 the foreign and retail predictions track the regime turns of the actual series (smoothed correlations 0.69 and

0.76) with a smaller amplitude due to L_1 shrinkage, while the institutional prediction is essentially pinned to zero (0.02).

Table 1 shows two things. First, dispersion across splits is large, so types cannot be compared from a single split. Second, **the naive persistence baseline beats SPR-IRL on directional accuracy and correlation**. Since a_t is published after the day- t close this is an informationally feasible rule. The reason is clear: the action has autocorrelation 0.401/0.149/0.345, yet the feature set contains no lagged own action. Restricting the features to (F1)–(F3) was a design choice ensuring that all types share only features corresponding to independently established regularities, not a design for predictive performance. Table 1 shows that the model has not merely fitted noise; it is not evidence of predictive superiority.

Because the saturation rate is zero for every type and split, the premise of Proposition 1.1 holds and the estimator is literally a Lasso.

2.3 Recovered Preferences: A Mirror Image

The mirror image on momentum. The foreign momentum weight is +0.0496 and positive in all 45 splits; the retail weight is −0.0304 and negative in all 45. Their bootstrap intervals do not overlap and each excludes zero (Figure 2A). Although the same feature set was estimated independently for each type, the two diverge in opposite directions, matching the sign expectations of prior work [1, 3].

Reappearance in the context response. The same mirror appears in the main effects (Figure 2B). Foreign investors move toward net buying when the market rises (+0.0361) and toward net selling when the won is weak (−0.0261); retail investors do the opposite on both (−0.0246, +0.0298). All four keep their sign in all 45 splits and all four intervals exclude zero.

Herding. The herding coefficient is negative in all 45 splits for all three types, and the foreign (−0.0402) and retail (−0.0322) intervals exclude zero. This does not separate a behavioural interpretation from the market-clearing constraint: under the −0.77 same-day correlation of §2.1, much of the negative coefficient may be mechanical. We therefore restrict the result to the descriptive statement that observed aggregate flow shows a negative cross-type association, and the same reservation applies to the mirrored main effects.

Underwater. The retail underwater weight is +0.0293 and positive in all 45 splits, consistent with the reference-price dependence of the disposition-effect literature [7, 8]. However, as Figure 2A shows, the interval contains zero for all three types. The retail interval is [−0.0122, +0.0665]: although the sign is 100% consistent across splits, it flips in 5.5% of month-block resamples. This is precisely the situation anticipated in §1.4, where sign consistency understates uncertainty because splits share observations. We therefore report a sign consistent with the disposition effect without claiming it as established.

Of the twelve interaction coefficients, four keep their sign in all 45 splits (foreign momentum $\times$ KOSPI200 −0.0139; retail momentum $\times$ FX +0.0144; retail herding $\times$ KOSPI200 −0.0080; retail underwater $\times$ FX −0.0186), of which the first and last also pass the bootstrap. Since twelve were examined jointly, we present them as exploratory only.

2.4 Stability

(V2) The bootstrap splits the results three ways: the eight terms forming the momentum and context mirrors all exclude zero, the underwater weights do not, and no institutional term does (the hollow markers in Figure 2).

(V3) In Figure 3 performance fluctuates sharply: foreign directional accuracy moves between 0.43 and 0.85, averaging 0.610 ± 0.116 , and the correlation falls to 0.131 ± 0.162 , less than half the CPCV value of 0.278. Because CPCV can place a test fold ahead of a training fold while walk-forward cannot, we report this degradation as a generalization limit from the temporal structure. The coefficient paths, by contrast, do not fluctuate: foreign momentum is positive in all ten windows and retail negative in all ten, the two never cross, and the same mirror holds for both context main effects. The institutional paths hug the zero line.

(V4) Foreign and retail coefficients agree to four decimal places under L_1 , L_2 and no penalty, so the mirror is not an artefact of the penalty. Only the institutional coefficients move (herding −0.0076 $\rightarrow$ −0.0124, underwater −0.0000 $\rightarrow$ −0.0018).

2.5 Institutional Investors: A Converged Null

For institutional flow SPR-IRL finds no preference that is stably identified at daily frequency, and we report this as a result. The evidence is fivefold: (i) the prediction series degenerates to zero (Figure 1); (ii) excluding herding, the coefficients are about one fortieth of the other types (−0.0006, −0.0008); (iii) no term excludes zero in the bootstrap (Figure 2); (iv) the coefficients depend on the choice of penalty; and (v) momentum sign consistency is 40% of ten walk-forward windows and underwater 0%.

This null is not an optimization failure. By Proposition 1.1, Equation (6) is convex with a unique global optimum, and comparing the solution reached against the closed-form optimum of the same objective leaves an institutional gap of only 0.17% on average and 0.48% at most (0.21% for foreign, 0.92% for retail). All three types reach the attainable optimum; only for institutional investors do the coefficients there sit near zero. The data do not determine an institutional preference; the optimizer did not fail to find one.

Economically this reads naturally: the “institutional” category aggregates pension funds, investment trusts, insurers, private funds and securities firms, so flows from agents with different objectives offset one another.

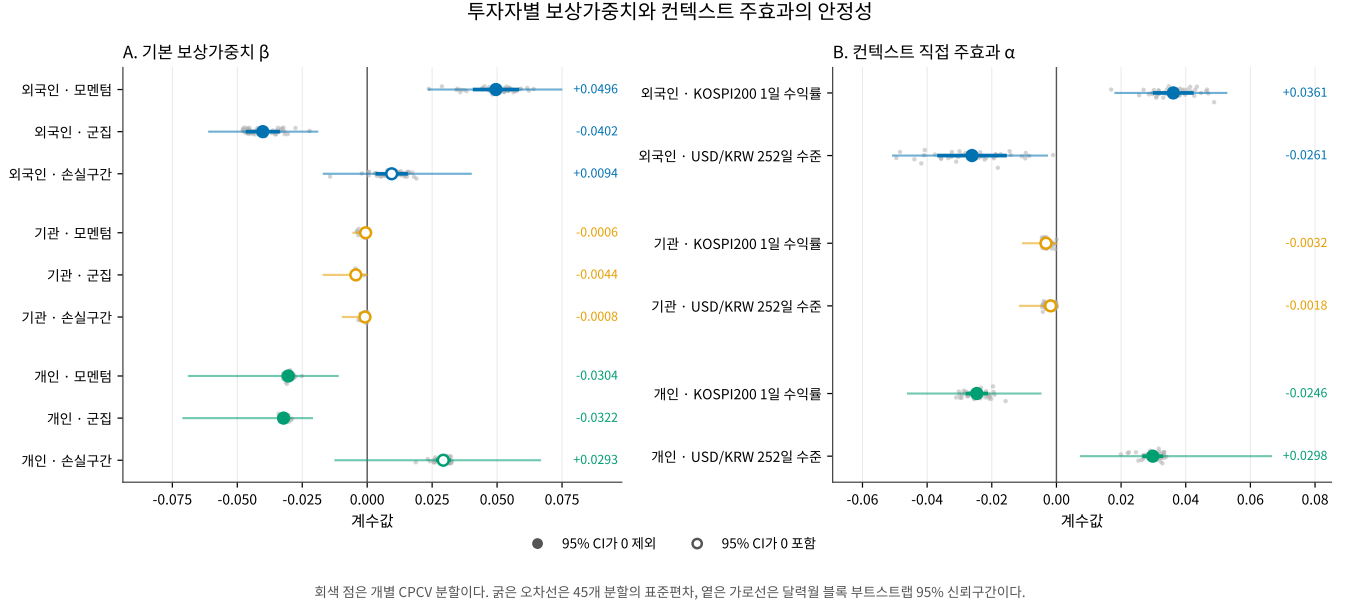


Figure 2: Stability of the recovered baseline reward weights β (A) and context main effects α (B). Grey dots are the 45 individual CPCV splits, thick error bars the across-split standard deviation, and light horizontal lines the 95% calendar-month block bootstrap interval. Hollow markers denote terms whose interval contains zero.

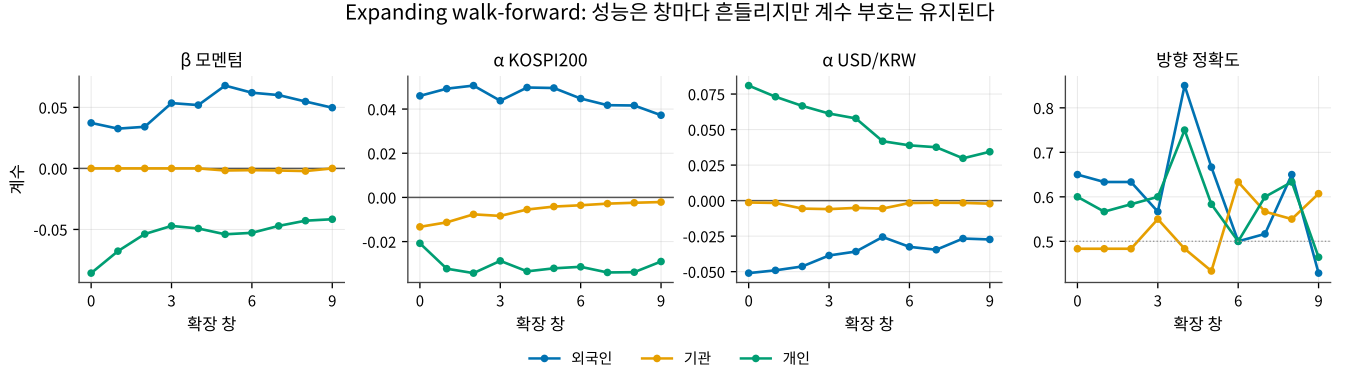


Figure 3: Coefficient paths and directional accuracy over ten expanding walk-forward windows. Performance fluctuates while the signs hold, and the foreign and retail momentum paths never cross.

2.6 Construct Validity

Table 2 reports the contrast. That the recovered signs agree with independently established regularities, and that two types diverge in opposite directions on momentum, is evidence of construct validity given that all types shared the same features and model form. The mismatch arises because the measured object differs: Lakonishok et al. [6] measure herding *within* the institutional group, whereas Equation (2) measures whether institutions move with the *other two types*, so our result is not a refutation of that literature.

3 Discussion and Limitations

What SPR-IRL recovers is a picture that places the preferences of three investor types side by side on one common scale. Foreign and retail investors diverge in opposite directions on momentum, the same contrast reappears in the context response, and institutional investors show no identified preference once herding is set aside. All three findings come from giving every type the same features and the same model form and estimating only the parameters separately, so the contrast was not imposed by design.

Table 2: Prior sign expectations and recovered results

Regularity	Expected	Recovered	Verdict
Foreign trend trading [1, 3]	$\beta^{mom} > 0$	+0.0496 (100%, CI excl. 0)	matches
Retail contrarian [3, 4]	$\beta^{mom} < 0$	−0.0304 (100%, CI excl. 0)	matches
Disposition effect [7, 8]	$\beta^{uw} > 0$	+0.0293 (100%, CI incl. 0)	partial
Institutional herding [6]	$\beta^{herd} > 0$	−0.0044 (100%, CI incl. 0)	mismatch

3.1 What This Study Does Not Claim

As the final stage of the protocol we state what the results do not support.

(1) Return predictability. Our metrics are reproduction errors for the direction and intensity of order flow and include no returns, Sharpe ratios or transaction costs. No trading rule follows from Table 1.

(2) Superior flow prediction. The persistence baseline beats SPR-IRL on directional accuracy and correlation (§2.2). Preference recovery and prediction call for different feature sets. In a preliminary check, adding the lagged own action as a fourth feature lifts the model above the baseline (directional accuracy 0.646/0.569/0.629) but reverses the sign of the retail α^{KOSPI} and eliminates the underwater coefficient. **Some of the mirror images reported here are therefore not separable from the autocorrelation of order flow.**

(3) Causal effects. The recovered weights are conditional associations among standardized observed features. Because feature–context correlations range from −0.29 to 0.22, the magnitude of any coefficient depends on the other terms being controlled for.

(4) A behavioural reading of the negative cross-type association. The herding coefficient and the mirrored context responses are not separable from the market-clearing constraint (§2.3).

(5) A test of the disposition effect. The retail underwater sign agrees with the literature but does not pass the month-block bootstrap, and the average-cost proxy is built from aggregate trades rather than account-level realized gains.

(6) Generalization across stocks. The single-stock design is a control for isolating heterogeneity across types; extension to other stocks and markets is untested.

3.2 Future Work

The most pressing task is to add the lagged own action as a feature and check whether the mirror images survive once autocorrelation is controlled. Next, decomposing the “institutional” category into pension funds, investment trusts, insurers and others may reveal preferences currently hidden by offsetting flows. Finally, the myopia assumption remains to

be relaxed. As Proposition 1.1 shows, a one-step-ahead setting makes the reward coincide with the policy; introducing state dynamics and multi-period value separates the two and turns the exercise into a structural recovery that regression cannot reach. The symmetric estimation and validation protocol of SPR-IRL carry over unchanged to that extension.

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